Inseok Kong

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Research Interests

I am broadly interested in **Trustworthy Artificial Intelligence**, with a primary focus on **Computer Vision**. My research has primarily centered on **Test-Time Adaptation (TTA)** to enable vision models to reliably adapt to real-world distribution shifts and environmental corruptions. Recently, I am expanding my research focus toward **Uncertainty Calibration** and object hallucination mitigation in vision-language models, striving to build reliable, well-calibrated, and robust visual intelligence systems.

Education

University of Seoul — *Mar 2022 – Present*
B.S. in GeoInformatics and Artificial Intelligence (Double Major) — Seoul, South Korea

- **Cumulative GPA:** 3.8 / 4.5
- **Relevant Coursework:** Machine Learning, Deep Learning, Pattern Recognition, Data Structures, Algorithms.

Publications & Preprints

[C: Conference, S: Submitted]

[C.1] **APC: Transferable and Efficient Adversarial Point Counterattack for Robust 3D Point Cloud Recognition** — *Jun 2026*
Geunyoung Jung, Soohong Kim, **Inseok Kong**, and Jiyoung Jung — *Denver, CO, USA*
IEEE/CVF Conference on Computer Vision and Pattern Recognition Findings (CVPR Findings)

[C.2] **MAMVI: 3D Test-Time Adaptation via Masked Multi-View Point Clouds** — *Aug 2026*
Inseok Kong, Geunyoung Jung, and Jiyoung Jung — *Lyon, France*
IEEE/IAPR International Conference on Pattern Recognition (ICPR) **(Oral Presentation)**

[S.1] **SATOR: Backpropagation-Free 3D Test-Time Adaptation via Sample-Adaptive Token Routing** — *Submitted*
Inseok Kong, Geunyoung Jung, and Jiyoung Jung
Under Review

[S.2] **Mitigating Feature Collapse for Test-Time Adaptation of 3D VLMs via Distribution-Guided Regularization** — *Submitted*
Geunyoung Jung, **Inseok Kong**, and Jiyoung Jung
Under Review

Research Experience

Statistical Inference and Information Theory Lab (SIIT) — *Jul 2026 – Present*
Undergraduate Research Student (Advisor: Prof. Junmo Kim) — KAIST, Daejeon, South Korea

- Conducting individual research on **Large Vision-Language Models (LVLMs)**, focusing on mitigating object hallucination and improving uncertainty calibration to address visual-textual misalignment and enhance multimodal trustworthiness.

Robotics and Computer Vision Lab (RCV) — *Jul 2025 – Jun 2026*
Undergraduate Research Student (Advisor: Prof. Jiyoung Jung) — University of Seoul, South Korea

- Researched **3D Test-Time Adaptation (TTA)** under severe domain shifts, investigating lightweight online adaptation strategies for 3D point cloud networks to maintain high inference accuracy while eliminating backpropagation overheads.
- Investigated **Point Cloud Robustness & 3D Adversarial Defense**, studying structural vulnerabilities of 3D point cloud classification architectures and exploring input-level defense mechanisms to neutralize point perturbations dynamically.
- Explored robust representation alignment for **3D Vision-Language Models (VLMs)**, addressing feature collapse and cross-modal misalignment issues under environmental corruptions and out-of-distribution (OOD) scenarios.

Projects

LG Aimers 8th Cohort — *Jan 2026 – Feb 2026*
Track: LLM Compression — Online

- Researched and applied model compression & optimization techniques (pruning, quantization) on EXAONE Large Language Models.

Optimal Autonomous Shuttle Route Planning *Oct 2025 – Jan 2026*
Capstone Design Project & Seoul Big Data Campus Seoul, South Korea

- Developed optimal route planning algorithms using mobility big data to support customized autonomous shuttle services for transportation-vulnerable elderly citizens in Seoul.

Sportier (UOS Entrepreneurship Club) *Mar 2025 – Jul 2025*
Lead Developer & President (Tools: React.js, JavaScript) Seoul, South Korea

- Led the entrepreneurship club as president and developed a sports matchmaking web application featuring location-based matching and competitive tier rankings.

Football Big Data Analytics *Feb 2025*
Football Big Data Camp Seoul, South Korea

- Investigated practical applications of LLMs and Retrieval-Augmented Generation (RAG) in football performance analytics.

Honors & Awards

Excellence Award *Jun 2026*
Seoul RISE AI Competition (Project: SATOR) Seoul, South Korea
Grand Prize (Seoul Mayor Award) *Oct 2025*
Seoul Metropolitan Big Data Campus Competition Seoul, South Korea
Excellence Award *Jan 2026*
UOS Capstone Design Project Seoul, South Korea
Encouragement Award *Aug 2025*
AICOSS Industry-Academia Hackathon (Diffusion & LLM Storybook) Seoul, South Korea
Dean's Award of College of Engineering *Jan 2025*
AICOSS Deep Learning Bootcamp (2D ViT Classification) Seoul, South Korea

Volunteer & Mentoring

KB Raschool 3rd Generation Mentor *Mar 2025 – Feb 2026*
Samsung Sori-Saem Welfare Center Seoul, South Korea

- Provided tailored mathematics mentoring and academic counseling for hearing-impaired 9th-grade students.

Seoul Donghaeng Project Mentor *Sep 2024 – Dec 2024*
Manna Regional Children's Center Seoul, South Korea

- Conducted mathematics mentoring and high school entrance guidance for multicultural 8th-grade students.

Skills, Certifications & Languages

- **Technical Skills:** Python, PyTorch, MATLAB, LaTeX, Git, Linux / Ubuntu
- **SQL Developer (SQLD)**, *Korea Data Agency (K-DATA)* *Dec 2024*
- **Advanced Data Analytics Semi-Professional (ADsP)**, *Korea Data Agency (K-DATA)* *Nov 2024*
- **English:** *TOEIC (835)* *Aug 2022 (Expired)*

Additional Information

- **Military Service:** Republic of Korea Army (Sergeant, Honorably Discharged) *Feb 2023 – Aug 2024*
- **Interests:** Formula 1 (Red Bull Racing / Max Verstappen)

Last updated: 2026-08-04